Day 1 TCP/IP

Task 1:- Open Wireshark, capture network traffic while you refresh the Flipkart homepage, and identify one packet each belonging to the Application, Transport, Internet, and Network Access layers by examining their headers.

Wireshark Practical: Flipkart Homepage

Step 1: Open Wireshark and select your active Wi-Fi interface. Click Start Capture.

Step 2: Open `https://www.flipkart.com` in your browser and refresh the homepage. Stop the capture.

Step 3: Identify these packets

TCP/IP Layer	Protocol	Header / Information
Application	DNS	Domain name query for Flipkart
Transport	TCP	Source port, destination port
Internet	IP	Source IP, destination IP
Network Access	Ethernet	Source MAC, destination MAC

Task 2:- Create a table comparing the OSI and TCP/IP models, listing all layers side-by-side, and for each layer, give a real-world example from any app you use (e.g., WhatsApp message, Zomato order).

Comparison of OSI and TCP/IP Models

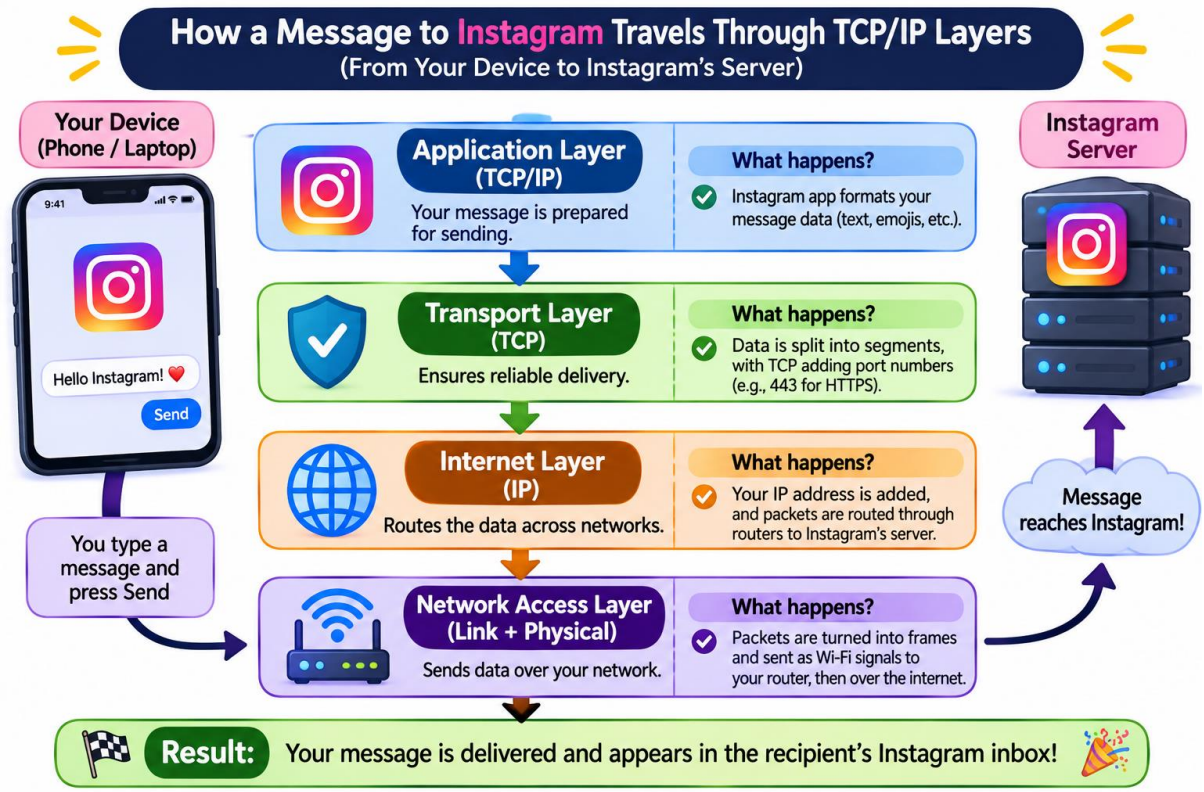
OSI Model (7 Layers)	TCP/IP Model (4 Layers)	Real-World Example
7. Application	Application	Sending a WhatsApp message
6. Presentation	Application	Encrypting WhatsApp messages
5. Session	Application	Maintaining a video call
4. Transport	Transport	TCP delivers data reliably
3. Network	Internet	IP routes a Zomato order request
2. Data Link	Network Access	Wi-Fi sends data from phone to router
1. Physical	Network Access	Radio signals carry data over Wi-Fi

Task 3:- Given a screenshot of a TCP segment header (provided by your trainer), label the fields that belong to the Transport layer and explain what each field does in 1-2 lines. Look for fields like Source Port, Destination Port, Sequence Number, etc.

TCP Segment Header – Transport Layer

Field	Function
Source Port	Identifies the sending application's port number.
Destination Port	Identifies the receiving application's port number.
Sequence Number	Identifies the position of data bytes in the TCP stream.
Acknowledgment Number	Indicates the next byte expected from the sender.
Header Length	Specifies the size of the TCP header.
Flags	Control connections and data transfer, such as SYN, ACK, and FIN.
Window Size	Indicates how much data the receiver can accept.
Checksum	Detects errors in the TCP header and data.
Urgent Pointer	Indicates urgent data when the URG flag is set.
Options	Carries additional TCP settings, such as maximum segment size (MSS).

Task 4:- Draw a diagram showing how a message sent from your device to Instagram's server passes through each TCP/IP layer, and annotate what happens at each step using simple app-related terms (e.g., 'message split into packets', 'IP address added', etc.).



Priyansh Bhaliya

Task 5:- Pick any one packet you captured in Wireshark and write which TCP/IP layer it belongs to, how you identified it, and what real-world action in an app (like sending a WhatsApp message or booking a ticket on BookMyShow) it represents.

Packet Analysis in Wireshark

- Protocol: TCP
- TCP/IP Layer: Transport Layer
- Identification: Identified by the `TCP` name in Wireshark's Protocol column and TCP header fields such as source port and destination port.
- Real-world Action: When I open Instagram, TCP helps transport data reliably between my device and the server.